

Multivariate Correlation

$X = (x_1, \dots, x_n)^T$ Joint Distribution $\longrightarrow$ 一般不明

$\hookrightarrow$ Correlation Matrix

$$R_X = E(X X^T), \quad R_X(i, j) = E(x_i x_j), \quad R_X = R_X^T$$

从三个角度认识 R_X :

① Decorrelation (Whitening 白化) $A \in \mathbb{R}^{n \times n}$, $Y = AX \in \mathbb{R}^n$

$$E(Y_i Y_j) = 0 \quad (i \neq j) \implies R_Y = \text{Diag} \longrightarrow \text{即正交化}$$

Goal: 寻找满足条件的矩阵 A .

$$R_Y = E[(AX)(AX)^T] = A E(X X^T) A^T = A R_X A^T \longrightarrow \text{对称矩阵对角化}$$

特征分解: $R_X = \sum_{k=1}^n \lambda_k U_k U_k^T = U \cdot \Lambda \cdot U^T$, 且 $U U^T = U^T U = I$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_k \geq 0, \forall k$

$$\text{取 } A = U^T \text{ 即可, 则: } U^T R_X U = U^T U \Lambda U^T U = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n).$$

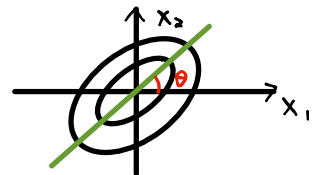
② Principal Component Analysis (PCA)

方向 $\alpha \in \mathbb{R}^n$, s.t. $E\| \text{Proj}_{\alpha} X \|^2 \longrightarrow$ 方差最大

$$= E \left\| \frac{\alpha^T X}{\alpha^T \alpha} \cdot \alpha \right\|^2$$

$$= \frac{E |\alpha^T X|^2}{\|\alpha\|^4} \cdot \|\alpha\|^2 = \frac{E |\alpha^T X|^2}{\|\alpha\|^2} = E \left(\frac{\alpha^T}{\|\alpha\|} X \right)^2$$

Dimensional Reduction
Compression



$$\text{优化问题: } \max_{\alpha} E\| \text{Proj}_{\alpha} X \|^2 = \max_{\alpha} E \left(\frac{\alpha^T}{\|\alpha\|} X \right)^2 \iff \max_{\alpha} E (\alpha^T X)^2, \text{ s.t. } \|\alpha\| = 1$$

$$E (\alpha^T X)^2 = \alpha^T E(X X^T) \alpha = \alpha^T R_X \alpha$$

因此优化问题可以写为: $\max_{\alpha} \alpha^T R_X \alpha$, s.t. $\|\alpha\| = 1$

利用 Lagrange, $L(\alpha, \lambda) = \alpha^T R_X \alpha - \lambda (\|\alpha\|^2 - 1)$

$$\nabla_{\alpha} L(\alpha, \lambda) = 2 R_X \alpha - 2 \lambda \alpha = 0 \implies R_X \alpha = \lambda \alpha \quad \alpha \text{ 即为特征向量}$$

代回: $\alpha^T R_X \alpha = \lambda \longrightarrow \alpha$ 对应的是最大特征值对应的特征向量

$\longrightarrow$ 角度 θ 与相关无关, 实际上与方差有关

$$E(x_1) = 0, \quad E(x_2) = 0, \quad E(x_1^2) = E(x_2^2) = 1, \quad E(x_1 x_2) = \rho, \quad X = (x_1, x_2)^T$$

$$R_X = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \implies \lambda_1 = 1 + \rho, \quad \lambda_2 = 1 - \rho \implies U_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

③ Expansion

$$Y = AX \implies X = A^T Y, \quad A = U^T, \quad \text{即 } X = U Y = \sum_{k=1}^n U_k \cdot Y_k$$

$$U_i^T U_j = 0 \quad (i \neq j) \quad \text{正交} \quad E(Y_i Y_j) = 0 \quad (i \neq j) \quad \text{正交}$$

Karhunen - Loeve Expansion (K-L 展开).

考虑 $X(t) = \sum_{k=-\infty}^{+\infty} \alpha_k \phi_k(t)$, 假设 $\{\phi_k\}$ 正交在区间 I 上, 且 $\int_I \phi_k^2(t) dt = 1$, $\int_I \phi_i(t) \phi_j(t) dt = 0 \quad (i \neq j)$

$$\text{则 } \alpha_k = \int_I X(t) \phi_k(t) dt$$

$$\begin{aligned}
 E(x_i x_j) &= E\left(\int_1 x(t) \phi_i(t) dt \int_1 x(s) \phi_j(s) ds\right) \\
 &= \int_1 \int_1 E(x(t) x(s)) \phi_i(t) \phi_j(s) dt ds \\
 &= \int_1 \int_1 R_x(t, s) \phi_i(t) \phi_j(s) dt ds \quad \rightarrow \text{我们希望为0 (i \neq j)}
 \end{aligned}$$

$$\Rightarrow \sum_m \sum_n R_x(m, n) \phi_i(m) \phi_j(n) \quad \checkmark \text{矩阵化}$$

$$\phi_i^T R_x \phi_j = 0 \quad \text{中 } \phi_k \text{ 为特征向量恰好满足}$$

回到连续情形: 需满足 $\int_1 R_x(t, s) \phi_i(s) ds = \lambda_i \phi_i(t)$, $\forall i$ (Integral Equation)

$$\text{取 } I = [-\frac{T}{2}, \frac{T}{2}], \quad \int_{-T/2}^{T/2} R_x(t-s) \phi_k(s) ds = \lambda_k \phi_k(t)$$

$$\rightarrow \text{给出一个解: } \phi_k(t) = \exp\left(j \frac{2k\pi}{T} t\right)$$

Prop: w.s.s. 且 $R_x(\tau) = R_x(\tau+T)$, 则:

$$x(t) = \sum_{k=-\infty}^{+\infty} \alpha_k \exp\left(j \frac{2k\pi}{T} t\right) \quad \text{为积分方程的解.}$$

$\rightarrow$ Bi-Orthogonal

$$\text{Stieltjes Integration: } \int f(x) dg(x) \Leftarrow \sum_k f(x_k) \Delta g(x_k)$$

Thm: (Spectral Representation)

$$\text{当 } T \rightarrow \infty, \text{ 有 } x(t) = \int_{-\infty}^{+\infty} \exp(j\omega t) dF_x(\omega)$$

$$\text{正交性的体现: } E\left(dF_x(\omega_1) \overline{dF_x(\omega_2)}\right) = 0$$

$$\begin{aligned}
 R_x(t, s) &= R_x(t-s) = E(x(t) \overline{x(s)}) \\
 &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \exp(j(\omega_1 t - \omega_2 s)) E\left(dF_x(\omega_1) \overline{dF_x(\omega_2)}\right) \\
 &= \int_{-\infty}^{+\infty} \exp(j\omega(t-s)) E|dF_x(\omega)|^2 \quad \leftarrow \\
 &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \exp(j\omega(t-s)) S_x(\omega) d\omega
 \end{aligned}$$

$$\text{因此, 我们有: } E|dF_x(\omega)|^2 = \frac{1}{2\pi} S_x(\omega) d\omega$$

$$\text{考虑: } Y(t) = \int_{-\infty}^{+\infty} h(t-\tau) x(\tau) d\tau, \quad x(\tau) = \int_{-\infty}^{+\infty} \exp(j\omega\tau) dF_x(\omega)$$

$$\begin{aligned}
 Y(t) &= \int_{-\infty}^{+\infty} h(t-\tau) \left(\int_{-\infty}^{+\infty} \exp(j\omega\tau) dF_x(\omega) \right) d\tau \\
 &= \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} h(t-\tau) \exp(j\omega(t-\tau)) d\tau \right) dF_x(\omega) \\
 &= \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} h(\tau') \exp(j\omega\tau') d\tau' \right) \exp(j\omega t) dF_x(\omega)
 \end{aligned}$$

$$= \int_{-\infty}^{+\infty} \exp(j\omega t) H(\omega) dF_x(\omega) = \int_{-\infty}^{+\infty} \exp(j\omega t) dF_Y(\omega)$$

$$\text{故有: } F_Y(\omega) = H(\omega) dF_x(\omega)$$

宽平稳 $x(t)$ 与 $\exp(j\omega t)$ 存在对应, $x(t) \longleftrightarrow \exp(j\omega t)$ 等距同构

$H_1 \quad H_2$

$$\text{即: } \|x(t) - x(s)\|_{H_1}^2 = \|\exp(j\omega t) - \exp(j\omega s)\|_{H_2}^2$$

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$$E|x(t) - x(s)|^2 = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_x(\omega) |\exp(j\omega t) - \exp(j\omega s)|^2 d\omega$$

Pf : 左边 = $2 R_x(0) - R_x(t-s) - R_x(s-t)$

$$\text{右边} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_x(\omega) \left[2 - \exp(j\omega(t-s)) - \exp(j\omega(s-t)) \right] d\omega$$

$$\text{而 } R_x(t-s) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S_x(\omega) \exp(j\omega(t-s)) d\omega$$

故 左边 = 右边

□